Individual dFNC

↓

Flatten the edges of each windows of every dFNC matrix into 1D Vector, and concatenate them to form a large matrix

↓

$$\begin{bmatrix} \begin{bmatrix} e_{1,1} & \cdots & e_{1,c} \\ \vdots & \ddots & \vdots \\ e_{w,1} & \cdots & e_{w,c} \end{bmatrix} \\ \begin{bmatrix} e_{w+1,1} & \cdots & e_{w+1,c} \\ \vdots & \ddots & \vdots \\ e_{2w,1} & \cdots & e_{2w,c} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} e_{(n-1)w,1} & \cdots & e_{(n-1)w,c} \\ \vdots & \ddots & \vdots \\ e_{nw,1} & \cdots & e_{nw,c} \end{bmatrix} \end{bmatrix}_{NW \times C}$$

where *n , w, and c* are respectively the number of subjects minus one, windows, and edges (upper triangle elements) in each dFNC correlation matrix.

Elbow ⇢ Clustering

↓

Centroids

↓

Compute minimum distances from centroids to label each window/ row of the concatenated matrix to form sequence of states across dFNCs

↓

labeled-frames ← 4